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CO2 emission prediction using machine and deep learning techniques

SUBJECT NAME:
Mathematics for computing 2 and
Elements of computing -2

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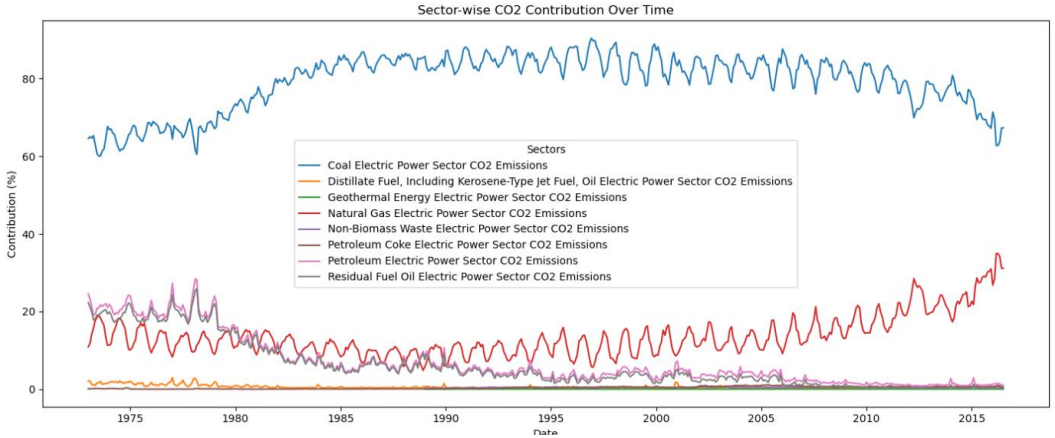
ABSTRACT

The rapid growth of computational technologies has led to increased energy consumption and carbon emissions, raising concerns about environmental sustainability. This project explores Green Computing by leveraging Deep Learning and Machine Learning techniques to predict CO_2 emissions across various sectors. Using a cleaned monthly sector-wise dataset, we implemented multiple predictive models, including Long Short Term Memory (LSTM), Ridge Regression, XGBoost, and LightGBM. The dataset was preprocessed through feature engineering, normalization, and lag feature generation, ensuring robust time-series forecasting. LSTM, a recurrent neural network, was used to capture temporal dependencies, while traditional regression, XGBoost, and LightGBM provided efficient and interpretable alternatives. The models were evaluated using performance metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score.Our results demonstrate the effectiveness of machine learning in forecasting CO_2 emissions, enabling policymakers and industries to make informed decisions for sustainable energy usage. This project highlights the role of AI-driven solutions in reducing environmental impact and promoting energy-efficient computing practices.

DATASET OVERVIEW

- The dataset contains 523 rows and 10 columns.
- The data spans from January 1973 onward.
- It records monthly CO_2 emissions from different energy sources in the electric power sector.
- **Columns**
  - Date,
  - Coal Electric Power Sector CO_2 Emissions,
  - Oil Electric Power Sector CO_2 Emissions,
  - Geothermal Energy Electric Power Sector CO_2 Emissions,
  - Natural Gas Electric Power Sector CO_2 Emissions,
  - Non-Biomass Waste Electric Power Sector CO_2 Emissions,
  - Petroleum Coke Electric Power Sector CO_2 Emissions,
  - Petroleum Electric Power Sector CO_2 Emissions,
  - Residual Fuel Oil Electric Power Sector CO_2 Emissions
  - Total Energy Electric Power Sector CO_2 Emissions
- **Source**
  - Kaggle

OBJECTIVE



- The primary objective of this project is to analyze and forecast monthly CO_2 emissions based on fuel-specific emissions data using advanced machine learning techniques. This includes:
- **Identifying key fuel contributors** to total CO_2 emissions.
- **Predicting total CO_2 emissions** from individual sectoral emissions (e.g., natural gas, coal, petroleum).
- **Forecasting future emissions trends** using temporal patterns in historical monthly data.
- **Comparing model performances** (MAE, RMSE, R^2) across multiple regression and time-series models such as Linear Regression, LSTM, XGBoost, LightGBM, and SVM.
- **Recommending optimal models** for accurate emission forecasting to aid in environmental planning and policy-making.

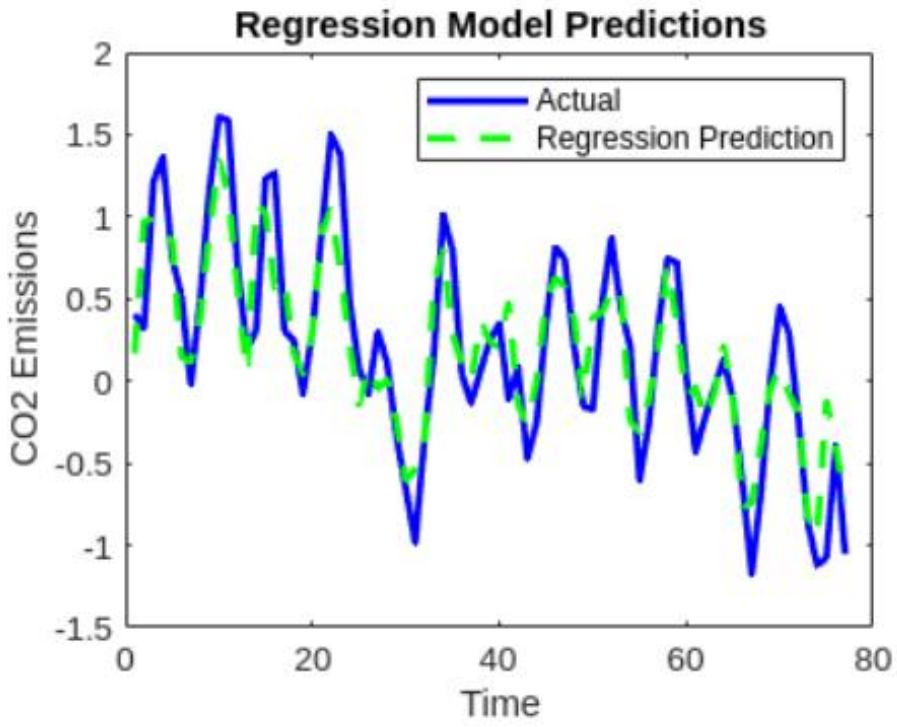
METHODOLOGY

- **Step 1: Data Collection**
  - Use monthly sectoral CO_2 emissions dataset from trusted sources (e.g., government or environmental databases).
- **Step 2: Data Preprocessing**
  - Handle missing values, outliers, and data inconsistencies.
  - Normalize or scale the features for better model performance.
  - Convert time-series date fields into usable formats (e.g., month, year index).
- **Step 3: Exploratory Data Analysis (EDA)**
  - Analyze trends, patterns, and seasonality in CO_2 emissions.
  - Visualize correlations between fuel-specific emissions and total emissions.
- **Step 4: Feature Engineering**
  - Create lag features to capture temporal dependencies.
  - Generate new features such as rolling averages or monthly year-on-year changes.
- **Step 5: Data Splitting**
  - Divide the dataset into training and testing sets (e.g., 85/15 or use TimeS eries Split).

- **Step 6: Model Selection & Training**
  - Apply multiple models:
    - **Ridge Regression**
    - **Support Vector Regression (SVR)**
    - **XGBoost Regressor**
    - **LightGBM Regressor**
    - **LSTM (for deep learning time-series forecasting)**
  - Tune hyperparameters for each model.
- **Step 7: Model Evaluation**
    - Use performance metrics:
      - **MAE** (Mean Absolute Error)
      - **RMSE** (Root Mean Squared Error)
      - **R^2 Score** (Coefficient of Determination)
- **Step 8: Forecasting**
  - Use the best-performing model to predict future CO_2 emissions.
  - Visualize actual vs. predicted emissions.
- **Step 9: Model Comparison**
  - Compare models based on accuracy, complexity, training time, and interpretability.
- **Step 10: Conclusion & Recommendations**
  - Summarize findings.
  - Suggest suitable models for real-time emission monitoring and policy implications.

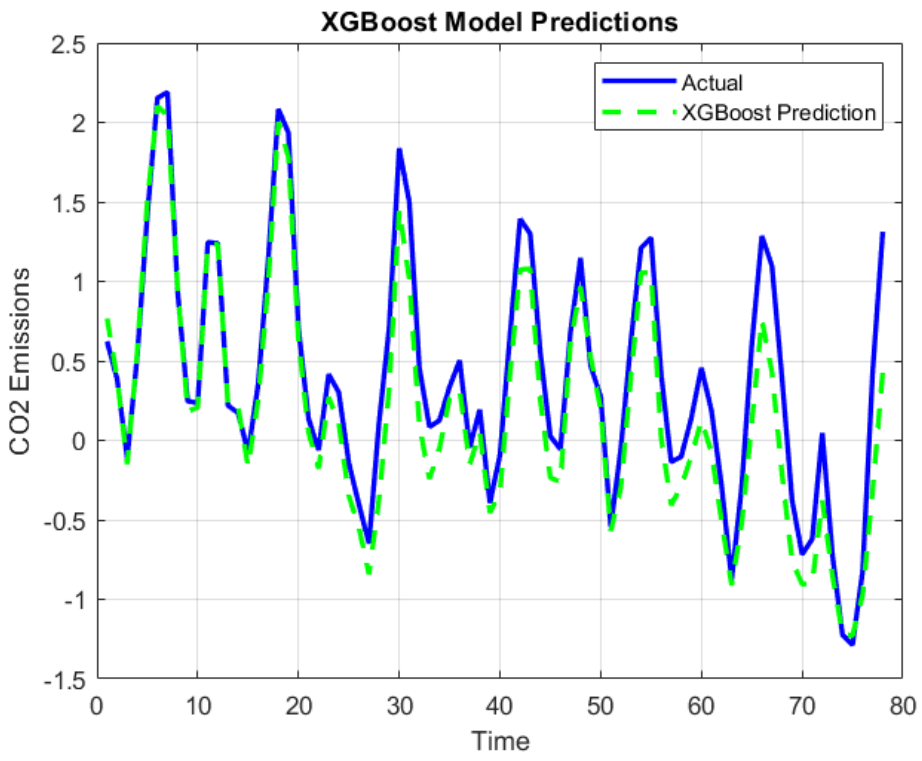
RIDGE REGRESSION

- A type of **linear regression** that includes **L2 regularization**.
- Helps **prevent overfitting** by penalizing large coefficients.
- Useful when features are **correlated** or the dataset has **multicollinearity**.
- Balances **bias-variance trade-off** using a regularization parameter (λ).
- Works best with **simple, linear relationships** in the data.
- Faster and interpretable but **less effective on non-linear patterns**.



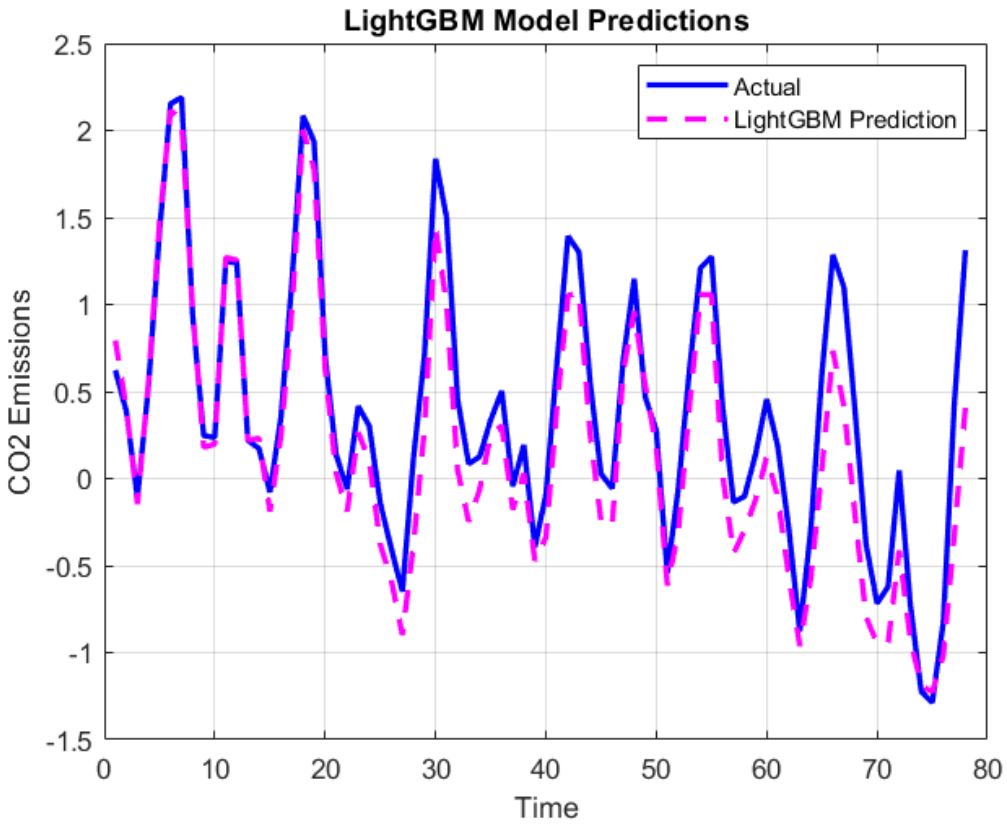
XG BOOST

- An **ensemble learning algorithm** based on decision trees.
- Boosts accuracy by combining **multiple weak learners**.
- Uses **gradient descent** to minimize errors over iterations.
- Includes built-in **regularization**, preventing overfitting.
- Handles **missing values and noisy data** effectively.
- Works well for **tabular and structured data**, including emissions datasets.
- Requires **feature scaling**, but is **highly efficient and scalable**.



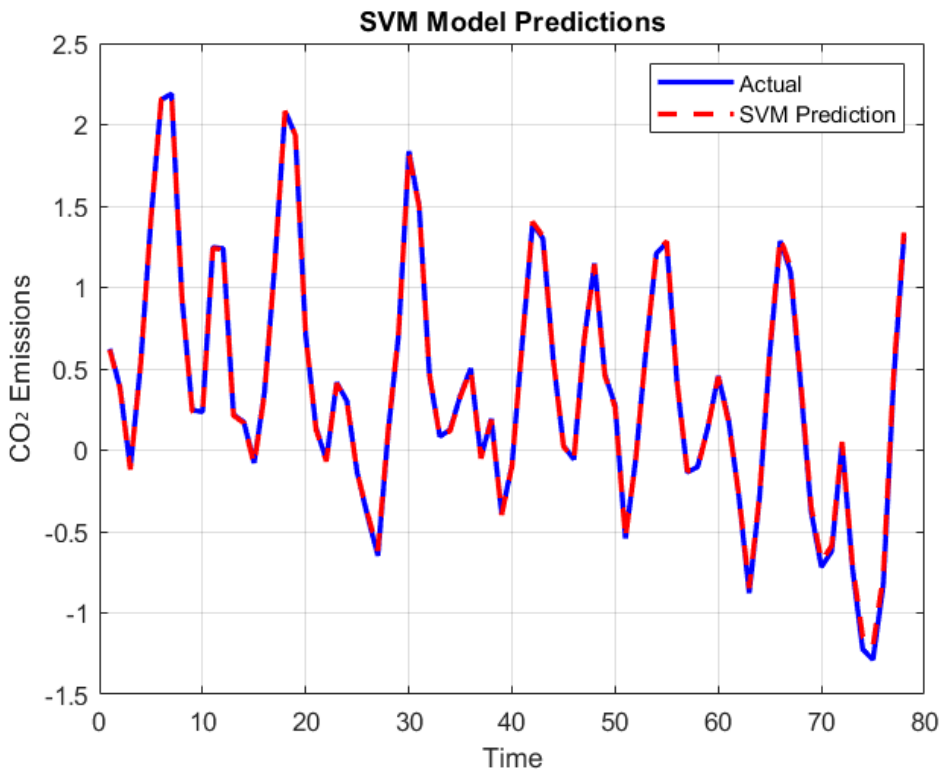
LIGHT GBM

- A **faster and more efficient alternative** to XGBoost.
- Uses **leaf-wise tree growth** for better accuracy.
- Optimized for **speed and low memory usage**.
- Supports **categorical features natively**.
- Best suited for **large-scale datasets** with many features.
- Can handle **complex, non-linear relationships** in CO_2 emissions data.
- May overfit on **small datasets**, so needs tuning.



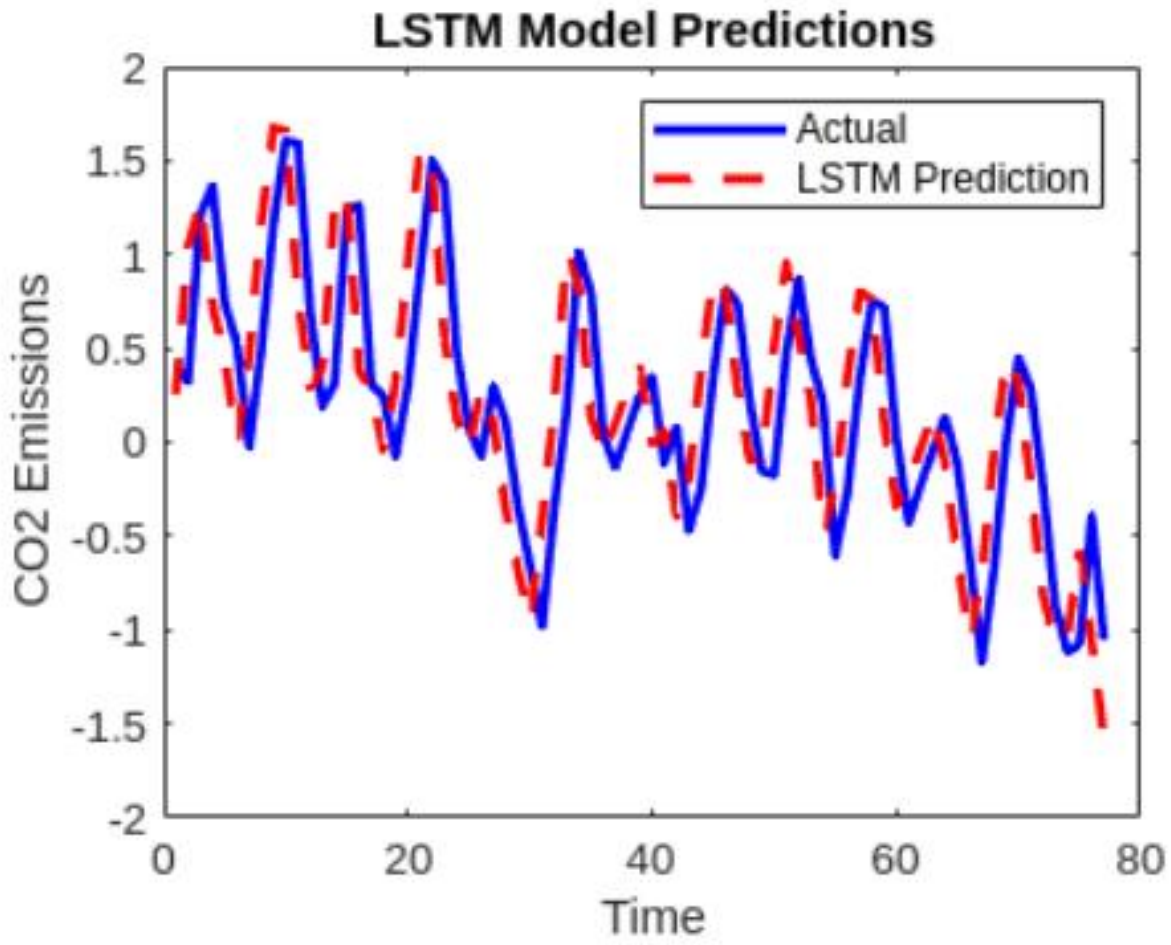
SVM(SUPPORT VECTOR MACHINE)

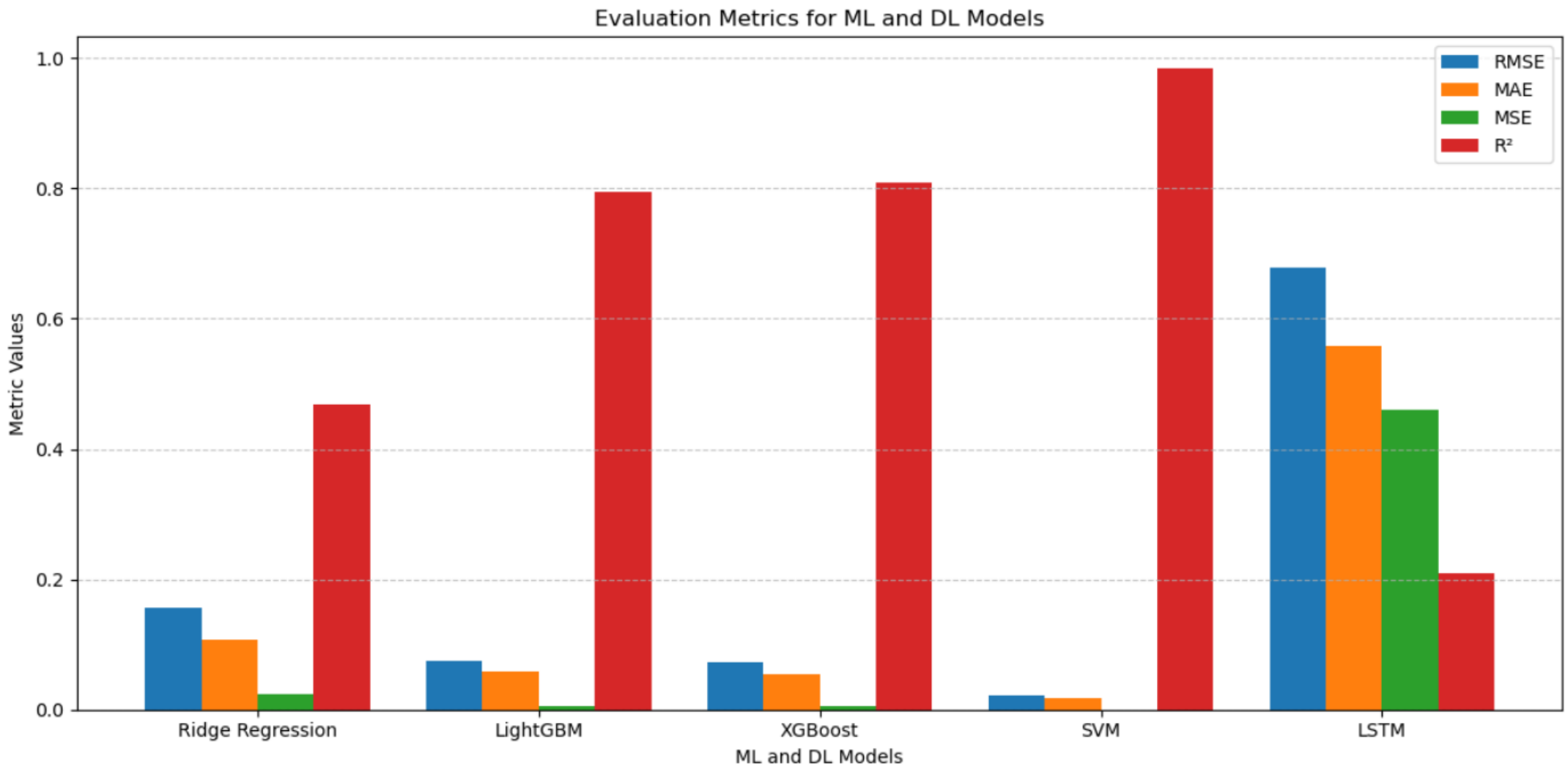
- A powerful supervised learning algorithm for **classification and regression.**
- Finds the **optimal hyperplane** that separates data points.
- Can be extended to **non-linear problems** using **kernel trick** (e.g., RBF).
- Effective in **high-dimensional spaces**, especially with fewer samples.
- **Robust to outliers**, especially with a good kernel choice.
- Requires **feature scaling** and **parameter tuning** (C, gamma).
- In emissions forecasting, SVM captures **non-linear temporal trends**.



LSTM MODEL

- A special type of **Recurrent Neural Network (RNN)** designed to handle **sequential/time series data**.
- Excels at learning **temporal dependencies** — trends and patterns over time.
- Uses **memory cells** to remember information for long periods.
- Solves the **vanishing gradient problem** found in traditional RNNs.
- Has **three gates**:
  - **Input gate** – controls what new information is stored.
  - **Forget gate** – decides what information to discard.
  - **Output gate** – selects what information to output.
- Performs well in modeling **non-linear, complex temporal relationships**.




Evaluation Metrics for ML and DL Models
RMSE
MAE
MSE
R²
Metric Values
1.0
0.8
0.6
0.4
0.2
0.0
Ridge Regression
LightGBM
XGBoost
SVM
LSTM
ML and DL Models

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize | DI4j Inference

Classifier

Choose **RandomForest** -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Num) Residual Fuel Oil Electric Power ...

Start | Stop

Result list (right-click for options)

17:25:04 - trees.RandomForest

Classifier output

```
=== Run information ===

Scheme:       weka.classifiers.trees.RandomForest -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1
Relation:     cleaned_monthly_sectoral_dataset-weka.filters.unsupervised.attribute.Remove-R1,10
Instances:    523
Attributes:   8
              Coal Electric Power Sector CO2 Emissions
              Distillate Fuel, Including Kerosene-Type Jet Fuel, Oil Electric Power Sector CO2 Emissions
              Geothermal Energy Electric Power Sector CO2 Emissions
              Natural Gas Electric Power Sector CO2 Emissions
              Non-Biomass Waste Electric Power Sector CO2 Emissions
              Petroleum Coke Electric Power Sector CO2 Emissions
              Petroleum Electric Power Sector CO2 Emissions
              Residual Fuel Oil Electric Power Sector CO2 Emissions
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

RandomForest

Bagging with 100 iterations and base learner

weka.classifiers.trees.RandomTree -K 0 -M 1.0 -V 0.001 -S 1 -do-not-check-capabilities

Time taken to build model: 0.36 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient                  0.9964
Mean absolute error                      0.3761
Root mean squared error                  0.6236
Relative absolute error                  6.9131 %
Root relative squared error              8.8413 %
Total Number of Instances              523
```

Status

OK

Log

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize | Dl4j Inference

Classifier

Choose **SimpleLinearRegression**

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Num) Residual Fuel Oil Electric Power Se...

Start | Stop

Result list (right-click for options)

- 17:25:04 - trees.RandomForest
- 17:25:33 - trees.RandomTree
- 17:25:58 - functions.SimpleLinearRegression

Classifier output

```
=== Run information ===

Scheme:       weka.classifiers.functions.SimpleLinearRegression
Relation:     cleaned_monthly_sectoral_dataset-weka.filters.unsupervised.attribute.Remove-R1,10
Instances:    523
Attributes:   8
              Coal Electric Power Sector CO2 Emissions
              Distillate Fuel, Including Kerosene-Type Jet Fuel, Oil Electric Power Sector CO2 Emissions
              Geothermal Energy Electric Power Sector CO2 Emissions
              Natural Gas Electric Power Sector CO2 Emissions
              Non-Biomass Waste Electric Power Sector CO2 Emissions
              Petroleum Coke Electric Power Sector CO2 Emissions
              Petroleum Electric Power Sector CO2 Emissions
              Residual Fuel Oil Electric Power Sector CO2 Emissions
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

Linear regression on Petroleum Electric Power Sector CO2 Emissions

0.98 * Petroleum Electric Power Sector CO2 Emissions - 1.26

Predicting 0 if attribute value is missing.


Time taken to build model: 0 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient                  0.9951
Mean absolute error                      0.5574
Root mean squared error                  0.694
Relative absolute error                 10.2452 %
Root relative squared error              9.8398 %
Total Number of Instances              523
```

Status

OK

Log

eka Explorer

ocess | Classify | Cluster | Associate | Select attributes | Visualize | Dl4j Inference

er

oose | **M5P** -M 4.0 -num-decimal-places 4

ptions

e training set

pplied test set | Set...

oss-validation | Folds | 10

rcentage split | % | 66

More options...

Residual Fuel Oil Electric Power Se...

Start | Stop

list (right-click for options)

04 - trees.RandomForest
33 - trees.RandomTree
58 - functions.SimpleLinearRegression
25 - trees.M5P

Classifier output

```
                Geothermal Energy Electric Power Sector CO2 Emissions
                Natural Gas Electric Power Sector CO2 Emissions
                Non-Biomass Waste Electric Power Sector CO2 Emissions
                Petroleum Coke Electric Power Sector CO2 Emissions
                Petroleum Electric Power Sector CO2 Emissions
                Residual Fuel Oil Electric Power Sector CO2 Emissions
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

M5 pruned model tree:
(using smoothed linear models)
LM1 (523/0.008%)

LM num: 1
Residual Fuel Oil Electric Power Sector CO2 Emissions =
        0 * Coal Electric Power Sector CO2 Emissions
        - 1 * Distillate Fuel, Including Kerosene-Type Jet Fuel, Oil Electric Power Sector CO2 Emissions
        + 0.0004 * Non-Biomass Waste Electric Power Sector CO2 Emissions
        - 1 * Petroleum Coke Electric Power Sector CO2 Emissions
        + 1 * Petroleum Electric Power Sector CO2 Emissions
        - 0.0006

Number of Rules : 1

Time taken to build model: 0.18 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient                  1
Mean absolute error                      0.0004
Root mean squared error                  0.0006
Relative absolute error                  0.0071 %
Root relative squared error              0.0085 %
Total Number of Instances              523
```

Log

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  An examination of daily CO_2 emissions prediction through a comparative analysis of machine learning, deep learning, and statistical models

- **Kumar M, Thakur S, Singh S**
  Machine learning-based time series models for effective CO_2 emission prediction in India

- **Ullah I, Qureshi AS, Kim D**
  Predicting CO_2 Emission Footprint Using AI through Machine Learning

- **Xing Liu**
  CO_2 emissions prediction based on regression, neural network and SVM

- **Zhao B, Li J, Zhang Y**
  Machine learning approaches for predictions of CO_2 emissions in the building sector

CONCLUSION

- This project focused on forecasting **CO_2 emissions** from the **electric power sector** using both **machine learning** and **deep learning** techniques.
- We trained and evaluated several models, including:
  - **Machine Learning Models:** Ridge Regression, Support Vector Machine (SVM), XGBoost, and LightGBM
  - **Deep Learning Model:** Long Short-Term Memory (LSTM)
- Each model was assessed using key evaluation metrics: **RMSE**, **R^2 Score**, and **MAE**.
- While deep learning showed promising results, particularly with LSTM capturing temporal dependencies, the **Support Vector Machine (SVM)** model outperformed others with the **highest accuracy and lowest error**.
- The results indicate that SVM is well-suited for handling complex, nonlinear relationships in emission data.
- These findings contribute to developing more accurate predictive tools for **climate analysis** and **energy policy planning**.